

3-5 Conservation of Mechanical Energy



Source / 來源

- 選修物理(二) 課堂講義 (4624R51) 教用版，第3章 功與能量，§ 3-5 力學能守恆，pp. 141–161
- Halliday, Resnick & Walker, *Fundamentals of Physics*, Ch 8: Potential Energy and Conservation of Energy
- Young & Freedman, *University Physics with Modern Physics*, Ch 8: Potential Energy and Conservation of Energy

Examples

Example 1 — Bowl and Projectile Motion (*p. 142*)

A small ball is released from rest at point A on a smooth spherical bowl of radius R . It leaves the bowl at a 60° opening B . Neglect resistance.

- (a) What motion follows after B ?
- (b) What is its speed at B ?
- (c) What maximum height above the ground does it reach if B is at height $R/2$?



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Solution

Try It 1 — Pegged Pendulum (*p. 142*)

A pendulum of length ℓ is released from rest at A . A peg at P makes the effective length after passing P equal to $\ell/2$. At B , the bob's string makes 60° with the vertical. Find its speed.

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 Solution

Example 2 — Vertical Circular Motion (*p. 144*)

A particle starts at A with speed $3\sqrt{gR}$ and enters the inside of a semicircular track of radius R . The horizontal distance AB is $3R$. Find:

- (a) The normal force at the top D .
- (b) The horizontal distance from B where it lands after leaving D .

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 Solution

Try It 2 — Losing Contact from a Circular Track (*p. 144*)

The same smooth circular track has radius R , but now $v_B = \sqrt{3gR} < \sqrt{5gR}$. Find:

- (a) The height where the particle leaves the track.
- (b) Its maximum height above the horizontal ground.

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 Solution

Example 3 — SHM Energy Ratio (*p. 147*)

A mass on a spring executes SHM with amplitude R and maximum speed v . When its kinetic and elastic potential energies have ratio 1 : 3, find:

- (a) Its speed in terms of v .
- (b) Its displacement magnitude from equilibrium.

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 Solution

Try It 3 — SHM Amplitude (*p. 147*)

A 1.0 kg mass is attached to a spring with $k = 0.16 \text{ N/cm}$ and moves on a smooth surface. When the spring is compressed 0.25 m, its speed is 2.0 m/s. Find the amplitude.

(A) 0.5 m (B) 0.75 m (C) 1.0 m (D) 1.25 m (E) 1.5 m

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 Solution

Example 4 — Escape Speed (*p. 149*)

Mars has radius 0.5 times Earth's and surface gravity 0.4 times Earth's. What is the ratio of escape speed from Mars to that from Earth?

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 Solution

Try It 4 — Maximum Height with Exact Gravity (*p. 149*)

A planet of radius R and surface gravity g launches a projectile vertically at $v_0 = \sqrt{gR}$. Find its maximum distance above the surface (no atmosphere).

(A) $R/4$ (B) $R/2$ (C) R (D) $3R/2$ (E) $2R$

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 Solution

Example 5 — Non-Conservative Work (*p. 150*)

A 60 kg person in a basin slides from rest down a 20 m high slope and reaches the bottom at 10 m/s. Find the work done by friction.

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 Solution

Try It 5 — Air-Resistance Work (*p. 150*)

A 0.20 kg ball is projected from height 20 m at speed 18 m/s and lands at 15 m/s. Find the work done by air resistance; take $g = 10 \text{ m/s}^2$.

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💡 Solution

Example 6 — Circular-Orbit Transfer (*p. 151*)

A satellite in a circular orbit of radius R has kinetic energy K . Friction reduces the system's mechanical energy by $K/4$, after which the satellite enters another circular orbit. What is its new kinetic energy?

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💡 Solution

Try It 6 — Dispersing a Binary System (*p. 151*)

Two equal masses m orbit their common center at radius r . What minimum energy is needed to separate them to infinity?

(A) $2Gm^2/r$ (B) Gm^2/r (C) $Gm^2/(2r)$ (D) $Gm^2/(4r)$ (E) $Gm^2/(8r)$

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 Solution

Quick Check

Ground-Level Gravity and Mechanical Energy

Q1. A mass m on a massless rod of length R is released from rest at 60° from the vertical. What is the rod tension at the lowest point?

(A) mg (B) $2mg$ (C) $3mg$ (D) $4mg$ (E) $5mg$

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 Solution

Q2. A pendulum of length R is released from horizontal. A peg a distance H below the pivot catches the string. What minimum H/R permits a complete circle about the peg?

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 Solution

Q3. In a smooth vertical circular track of radius R , a rider's maximum speed is $\sqrt{5}$ times the minimum speed. What is the normal-force G level at the lowest point?

(A) $4G$ (B) $5G$ (C) $6G$ (D) $7G$ (E) $8G$

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 Solution

Elastic Potential and SHM

Q4. A spring-mass oscillator has amplitude R . When $|x| = 2R/3$, what fraction of its maximum kinetic energy is kinetic energy?

(A) $1/3$ (B) $4/9$ (C) $5/9$ (D) $2/3$ (E) 1

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 Solution

Q5. A satellite in a circular orbit experiences small atmospheric drag and spirals inward. Which quantities decrease? (*Select 3.*)

(A) Mechanical energy (B) Orbital radius (C) Gravitational potential energy (D) Kinetic energy
(E) Centripetal acceleration

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 Solution

Q6. A satellite of mass m orbits Earth at radius $2R$. Express its kinetic energy and total mechanical energy using G, R, M, m .

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 Solution

Q7. A cyclist rides in still air. Static friction exceeds kinetic friction. Which statements are correct? (*Select 2.*)

(A) At higher constant speed, more effort is needed mainly because of air drag.

(B) The cyclist–bicycle system's momentum is conserved.

(C) The ground's normal force does no work on the cyclist–bicycle system.

(D) If the tire slips, kinetic friction does no work on the system.

(E) Including Earth, the system's mechanical energy is conserved during acceleration and deceleration.

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 Solution

Q8. A rocket receives energy E to rise from Earth's surface to $2r$ from the center and stop. What energy is needed to rise to $3r$ and stop?

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 Solution

Challenge

Ground-Level Gravity and Mechanical Energy

Q1. A bead is released above the opening A of a smooth circular tube of radius R . It exits at B and just re-enters A as a projectile. Find the launch speed at B .

 Solution

Q2. In the same setup, the first release height is h_1 ; a second release just reaches B with zero speed. Find h_1/h_2 .

 Solution

Q3. A person swings on a 20 m rope from platform A to a lower platform B , 4 m below. The rope can sustain 1120 N. What maximum body mass avoids breaking the rope? Take $g = 10 \text{ m/s}^2$.

(A) 40 kg (B) 60 kg (C) 80 kg (D) 100 kg (E) 120 kg

Spring–Mass System

Q4. A block of mass m approaches a spring with speed v on a frictionless surface. Which statements are correct? (*Select 3.*)

- (A) Maximum compression is $x = mv / \sqrt{mk}$.
- (B) Kinetic energy is conserved while compressing.
- (C) Speed on returning to the initial point is v .
- (D) The spring force is maximum at first contact.
- (E) The spring force is maximum at maximum compression.

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Q5. With kinetic friction present in the same problem, which statements are correct? (*Select 2.*)

- (A) Maximum compression is less than $v \sqrt{m/k}$.

- (B) Return speed at the initial point is less than v .
- (C) Return speed equals v .
- (D) Mechanical energy is minimum at maximum compression.
- (E) Mechanical energy is maximum at maximum compression.

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 Solution

Q6. A satellite's elliptical orbit has perigee $2R$ and apogee $4R$. Given Earth's surface gravity g , find its total mechanical energy.

- (A) $-mgR/2$ (B) $-mgR/3$ (C) $-mgR/4$ (D) $-mgR/6$ (E) $-mgR/8$

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 Solution

Q7. A satellite changes from a circular orbit of radius $3R$ (initial mechanical energy $E < 0$) to $2R$ because of atmospheric drag. Which statements are correct? (*Select 3.*)

- (A) At $3R$, its kinetic energy is $-E$.
- (B) At $2R$, its potential energy is $3E$.
- (C) Its speed is lower at $2R$.
- (D) Gravity does work $-E$ during the transfer.
- (E) Air resistance does work E during the transfer.

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 Solution

Orbital Transfer Group

Q8. A spacecraft moves in a circular orbit of radius $2R$ and changes to an ellipse tangent to Earth's surface at Q and tangent to the old orbit at P . Find its speed at P after the change.

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 Solution

Q9. What work must the engine do to make this return transfer?

 Solution

Q10. A probe's elliptical orbit has perigee $0.4R$ and apogee $1.6R$; its apogee kinetic energy is K . How much energy must the engine supply to move it to a circular orbit of radius $3R$?

 Solution

Exam Archive

Q1 — Camera Dropped from a Rising Balloon (*p. 158, 109 學測*)

A camera is released from height 100 m while a balloon rises at 5.0 m/s. Neglect air resistance and take $g = 10 \text{ m/s}^2$. What is the impact speed?

(A) 55 (B) 45 (C) 35 (D) 25 (E) 15 m/s

💡 Solution

Q2 — Spring Gun Recoil (*p. 158, 106 指考*)

A 1.0 kg spring gun on a frictionless floor fires a 0.20 kg steel ball at 4.0 m/s. The spring constant is 2.4×10^3 N/m. Find the initial compression.

(A) 3.6 cm (B) 4.0 cm (C) 4.6 cm (D) 5.0 cm (E) 5.6 cm

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💡 Solution

Q3 — Rough Incline and Stopping Distance (*p. 158, 106 指考*)

A mass m slides down a rough incline of length L at initial speed v_0 , angle θ , and coefficient μ ; it then slides on a rough horizontal floor until stopping. Find:

- (a) Work by friction on the incline.
- (b) Horizontal stopping distance d .
- (c) Time spent on the horizontal floor.

 Solution

Q4 — Basketball Mechanical-Energy Loss (*p. 158, 105 指考*)

A 0.600 kg basketball is thrown horizontally from height 100 m at 10 m/s and hits the ground at 40 m/s. Take $g = 10 \text{ m/s}^2$. How much mechanical energy is lost?

(A) 80 J (B) 150 J (C) 220 J (D) 300 J (E) 680 J

 Solution

Q5 — Kinetic Energy versus Height with Air Drag (*p. 158, 104 指考*)

A basketball is thrown upward at an angle and caught at a lower height. Which graph best represents kinetic energy K versus height h when air resistance is present?

[Figure: Five candidate K - h curves; the correct curve has smaller K on descent than on ascent at the same height.]

 Solution

Q6 — Energy to a Geosynchronous Orbit (*p. 158, 104 指考*)

A satellite circles Earth at radius R with period 180 min. What minimum energy moves it to a geosynchronous circular orbit? (A geosynchronous period is 1440 min.)

(A) $3GMm/(8R)$ (B) $GMm/(4R)$ (C) $GMm/(2R)$ (D) $5GMm/(8R)$ (E) $3GMm/(4R)$

 Solution

Q7 — Spring Energy and Kinetic Energy (*p. 159, 101 指考*)

Two blocks of masses M and m are connected by a spring on a frictionless table; m is fixed and M is pulled a distance ΔL and released. When the block's kinetic energy equals the spring potential energy, what is the extension?

(A) $\Delta L/2$ (B) $\Delta L/\sqrt{2}$ (C) ΔL (D) $\sqrt{2}\Delta L$ (E) $2\Delta L$

 Solution

Q8 — Atomic Force and Kinetic Energy (*p. 159, 101 指考*)

An atom starts from rest at P under a one-dimensional force shown in an $F-x$ graph. The force is positive from P to equilibrium point Q and negative to the right. Which statement is correct?

(A) Kinetic energy is maximum at Q .

(B) Kinetic energy is maximum at R .

(C) It stops permanently at Q .

(D) It stops permanently at R .

(E) Its velocity at R must point in $-x$.

 Solution

Q9 — Jumping Robot and Salmon (*p. 160–161*, 素養導向)

SALTO has mass 0.10 kg, spring constant 100 N/m, and takeoff speed 1.75 m/s. A bushbaby has vertical takeoff speed 2.2 m/s and jumps 5 times in 4 s. A salmon's maximum swimming speed is 2.8 m/s. Take $g = 10 \text{ m/s}^2$ for the robot and bushbaby, and $g = 9.8 \text{ m/s}^2$ for the salmon.

- (a) What spring deformation gives SALTO its takeoff speed?
- (b) If SALTO jumps at 2 Hz, compare its jump agility with that of the bushbaby.
- (c) Can the salmon clear a 4 m rapid using kinetic energy alone?

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 Solution

Glossary

English	中文	Symbol / Unit
mechanical energy	力學能	$E = E_k + U$ [J]
conservation of mechanical energy	力學能守恆	$E_i = E_f$
conservation of energy	能量守恆	$E_i + W_{nc} = E_f$
non-conservative work	非保守力作功	$W_{nc} = \Delta E$
escape speed	脫離速度	$v_e = \sqrt{2GM/r}$
escape energy	脫離能量	GMm/r
bound energy	束縛能	$-E$
circular orbit	圓軌道	$E = -GMm/(2r)$
elliptical orbit	橢圓軌道	$E = -GMm/(2a)$
semimajor axis	半長軸	a
orbital transfer	軌道變換	—
simple harmonic motion	簡諧運動	SHM
amplitude	振幅	R or A
spring constant	彈簧力常數	k [N/m]
elastic potential energy	彈性能	$U_s = \frac{1}{2}kx^2$
normal force	正向力	N
G-force	G 力	$N/(mg)$

air resistance / drag	空氣阻力	—
power amplification	力量放大	